

POWER BI PROJECT

INTRODUCTION:

Power BI is a powerful business intelligence tool developed by Microsoft that allows users to visualize, analyst, and share insights from data. In today's data-driven world, organizations rely on clear and interactive reports to make better decisions. Power BI makes it possible to transform raw data into visually appealing dashboards that provide actionable insights.

WHAT IS POWER BI?

Power BI is a data visualization and business analytics tool that connects to various data sources and helps users create interactive reports and dashboards. It combines data preparation, visualization, and sharing in one platform. Supply Chain Performance & Optimization Dashboard This may include:

- **Data Connectivity** – Importing or connecting live data from sources like Excel, SQL, or cloud platforms.
- **Data Transformation** – Cleaning, shaping, and modelling data using Power Query and DAX.
- **Data Visualization** – Creating interactive charts, graphs, maps, and KPI cards.
- **Dashboard Creation** – Building user-friendly dashboards that highlight key business metrics.
- **Data Sharing** – Publishing reports to the Power BI Service for collaboration and decision-making.

WHAT IS POWER QUERY EDITOR?

Power Query Editor is a data transformation tool in Power BI (also available in Excel). It allows users to connect, clean, and shape data before loading it into Power BI for analysis and visualization.

It provides an easy-to-use interface with point-and-click options as well as an underlying M language (Power Query Formula Language) for advanced transformations.

Key Features of Power Query Editor:

- **Data Import** – Connect to multiple sources like Excel, SQL, Web APIs, CSV, etc.
- **Data Cleaning** – Remove duplicates, handle missing values, filter out irrelevant rows, and correct errors.
- **Data Transformation** – Split columns, merge tables, pivot/unpiloted, change data types, and create custom columns.

Data Shaping – Reorganize data structures to make them analysis-ready.

- **Applied Steps** – Every transformation is recorded as a step, making the process transparent and reversible.
- **Automation** – Once defined, the same query can be refreshed automatically with updated data

INTRODUCTION TO THE ICECREAM PRODUCT DATASET

The dataset represents a comprehensive overview of supply chain operations, covering the flow of products from manufacturing to customer delivery. It contains 1000 records with 10 attributes, combining data on sales, inventory, suppliers, logistics, and quality management.

This dataset was sourced from the Kaggle website, making it publicly available for data analytics, business intelligence, and academic projects.

COLUMN DETAILS – ICECREAMS DATASETS

- **Product ID:** (Unique identifier for each ice cream product. (IC1000))
- **Flavour:** (The flavor or type of ice cream. (Cookies & Cream, Mango, Butterscotch))
- **Brand:** (The company or manufacturer of the ice cream. (Kuality Walls, Amul, Vadilal, Häagen-Dazs))
- **Category:** (The product type or packaging category. (Cone, Stick, Cup))
- **Size:** (The quantity or volume of the ice cream (in milliliters). (150, 200))
- **Prices:** (Price of the ice cream in Indian Rupees. (158.0, 499.0, 29.0))
- **Calories:** (Number of calories per serving. (138, 286, 375))
- **Production Date:** (The date the ice cream was manufactured. (2023-01-01))
- **Expiry Date:** (The date the ice cream expires. (2023-06-01))
- **Rating:** (Customer rating or quality score (likely out of 5). (3.5, 2.9))

DATA CLEANING & PREPARATION NOTE

The dataset was directly imported into Power BI from Kaggle. Normally, in ice cream sales Report analytics, the Power Query Editor is used to clean and pre-process the dataset (e.g., handling missing values, removing duplicates, renaming columns, fixing data types).

However, in this case:

- The dataset contained no missing values.
- There were no duplicates.
- Column data types were correctly assigned (numeric, categorical, etc.).
- Values were already in a clean and consistent format.

